Chi-square (χ^2) test

→ ul test whether the difference b/w what we expect and what we observe is due to a chance or not.

→ if observed values are very close to the expected then no significant difference.

→ if they differ a lot, then significance difference exist.

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

O → observed value

E → Expected value

Steps to solve chi square

1. → write H_0 & H_1
2. → find Expected frequency (E)
3. → Apply χ^2 formula
4. → find degree of freedom
5. → Compare calculated χ^2 with table value
6. → Decide : Reject or accept H_0

Q → A die is thrown 60 times. Outcomes are :—

Face	1	2	3	4	5	6
Observed (O)	8	10	9	11	12	10

test whether the die is fair at 5% level.

$H_0 \Rightarrow$ die is fair.

$H_1 \Rightarrow$ die is not fair.

total throws = 60

for a fair die

$$E = \frac{1}{6} \times 60 = 10$$

χ^2 calculation

Face	O	E	O - E	$(O-E)^2$	$(O-E)^2/E$
1	8	10	-2	4	0.4
2	10	10	0	0	0
3	9	10	-1	1	0.1
4	11	10	1	1	0.1
5	12	10	2	4	0.4
6	10	10	0	0	0
					1

$\chi^2 = 1$

degree of freedom = df = $n-1$ = $6-1 = 5$

n → no of categories / outcome

at 5% level $\chi^2_{0.05}(5) = 11.07$

since $1 < 11.07$ (accept H_0)

Conclusion is Die is fair.

Q → Relationship between gender & preferences for online classes

	Like	Dislike	total
Male	30	20	50
Female	20	30	50
total	50	50	100

test for independence at 5% level.

$H_0 \Rightarrow$ gender & preference are independent.

$H_1 \Rightarrow$ they are dependent.

$$E = \frac{(\text{Row total} * \text{column total})}{\text{grand total}}$$

for male - like :-

$$E = \frac{50 * 50}{100} = 25$$

M - D :- E = 25

F - L :- E = 25

F - D : E = 25

χ^2 table

Cells	O	E	$(O-E)^2/E$
M - like	30	25	1
M - dislike	20	25	1
F - like	20	25	1
F - dislike	30	25	1
			4

$x_1 + x_2 + x_3 = 10$

7 1

2

$\chi^2 = 4$

$df = (r-1) * (c-1) = (2-1)(2-1) = 1$

$\chi^2_{0.05}(1) = 3.84$

$4 > 3.84$ (Reject the H_0)

Conclusion is gender & preferences are related.

Q →

Gender	Mobile	Laptop	total
Male	35	15	50
Female	25	25	50
total	60	40	100

test whether gender & device preference are independent at 5% level.

$H_0 \Rightarrow$ gender & device preferences are independent

$H_1 \Rightarrow$ gender & device preferences are dependent

Now compute expected value for each cell :—

$$E = \frac{\text{Row total} * \text{column total}}{\text{grand total}}$$

$$E_{MM} = \frac{50 * 60}{100} = 30$$

$$E_{ML} = \frac{50 * 40}{100} = 20$$

$$E_{FM} = \frac{50 * 60}{100} = 30$$

$$E_{FL} = \frac{50 * 40}{100} = 20$$

Add all values

$$\chi^2 = 0.83 + 1.25 + 0.83 + 1.25 = 4.16$$

degree of freedom

$$df = (r-1) * (c-1)$$
$$= (2-1) * (2-1)$$
$$= 1$$

at 5% level and df 1

$$\chi^2_{0.05} = 3.84$$

$$\chi^2_{calculated} = 4.16$$

$$\chi^2_{table} = 3.84$$

Since $4.16 > 3.84$ (Reject H_0)